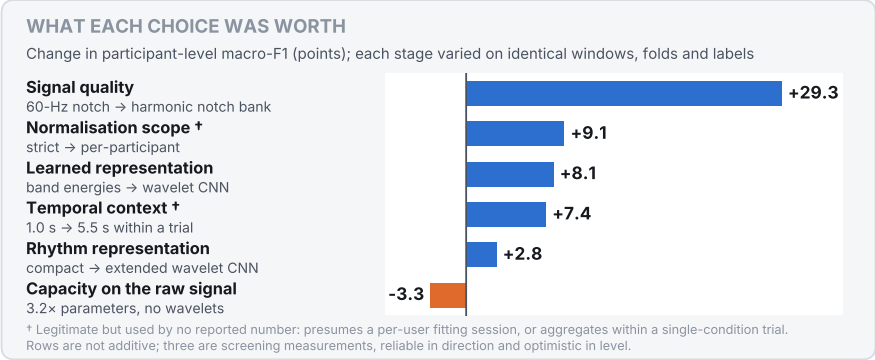
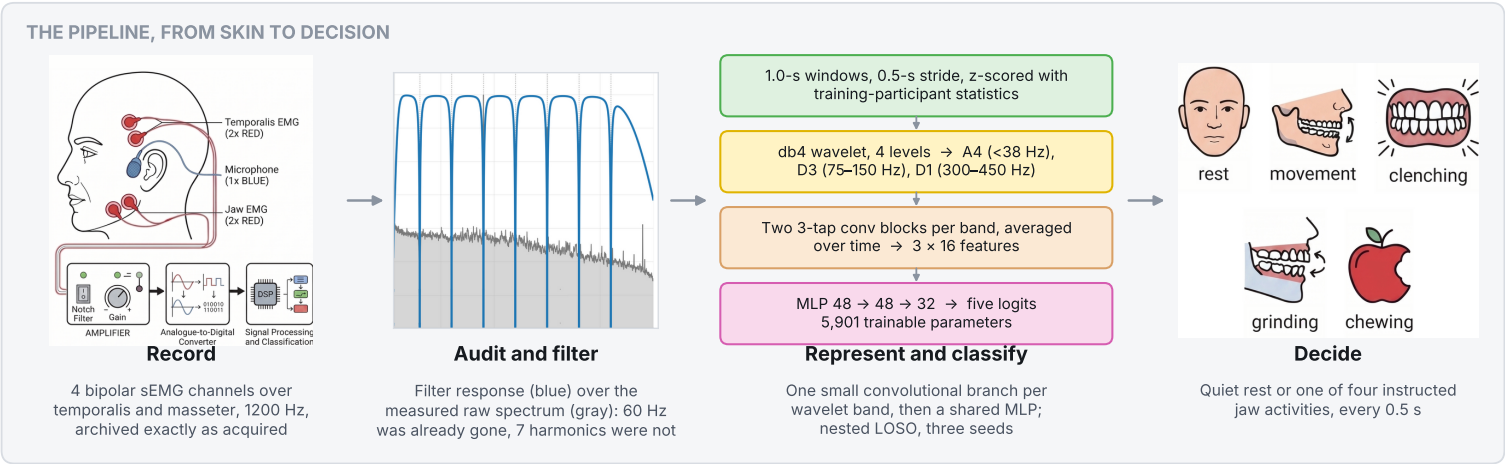


Signal Quality, Representation, and Pre-Modelling Choices

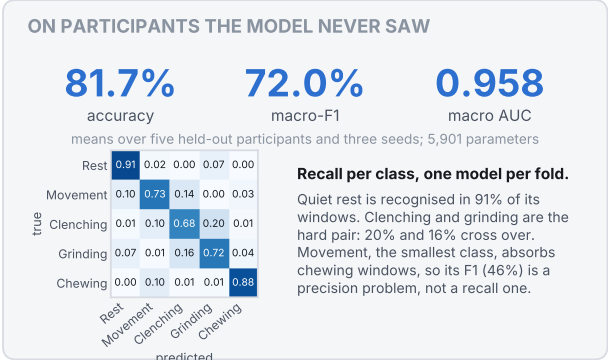
A measured performance budget for surface-EMG jaw-activity classification on unseen participants

5 adults with diagnosed bruxism · 100 one-minute recordings · 6,173 one-second windows · quiet rest and 4 instructed jaw activities · nested leave-one-subject-out validation



Signal quality set the ceiling
The best model on the uncorrected signal (43.5% macro-F1) scored below the weakest baseline on the corrected one (61.3%). Correcting the filter chain was worth 29.3 points.

Representation mattered more than size
Both wavelet models placed above both raw-signal models; 3.2× the parameters on the raw signal lost 3.3 points. Temporal context and normalisation scope were worth as much as the architecture or more.



What transfers is the procedure
Audit the signal against what the analysis assumes, publish the interference statistic beside the accuracy, and price the choices made before modelling against the model.

Signal Quality, Representation, and Pre-Modelling Choices: A Measured Performance Budget for Surface-EMG Jaw-Activity Classification on Unseen Participants

Mohammad Hassan Farhadi, Ali Rabiee, Rahul Singh, Mariusz P. Furmanek, Reza Abiri*, and Theodore A. Walls*

Abstract—Goal: Pipelines that classify physiological signals are reported through a final accuracy that says little about which stage earned it. We ask which stages of one surface-electromyography (EMG) pipeline account for its performance. **Methods:** One hundred recordings from five adults with diagnosed bruxism (quiet rest and four instructed jaw activities) were classified with a wavelet convolutional network under nested leave-one-subject-out validation. Each stage was then varied on identical windows, folds and labels, and its effect placed on one scale. **Results:** Unseen-participant performance was 81.7% accuracy and 72.0% macro-F1. Correcting a mains-harmonic defect that a conventional single-notch chain had missed was worth 29.3 points of macro-F1; normalisation scope 9.1, learned representation 8.1, temporal context 7.4; tripling parameters on the raw signal lost 3.3. **Conclusions:** Signal quality sets a ceiling, representation matters more than size, and choices fixed before modelling can outweigh the model. The procedure transfers to any biosignal pipeline.

Index Terms—Bruxism, convolutional neural networks, performance attribution, signal-quality audit, surface electromyography.

Impact Statement—Pricing each stage of an EMG classification pipeline on identical data showed signal quality, then representation, then pre-modelling choices outweighed model size: a procedure any biosignal study can apply.

I. INTRODUCTION

A classifier for physiological signals is the last stage of a long pipeline: a sensor, an amplifier with filters of its own, a filter chain, a segmentation rule, a normalisation scope and an evaluation protocol, any of which can move the reported number by more than the model. Yet pipelines in electromyography (EMG), electroencephalography (EEG),

electrocardiography and inertial sensing are reported almost entirely through a final accuracy and a block diagram [1], [2]. Seed and split variance often exceeds the gains papers claim [3], and reporting standards now ask for an attribution a single accuracy cannot supply [4], [5].

This paper is a stage-by-stage analysis of one such pipeline. We built one pipeline end to end, from the participant's chair to the decision, and priced it: each stage is varied with everything else fixed, on the same windows, folds and labels, and the effects placed on one scale. Its parts, a signal audit, a wavelet decomposition, a small convolutional network and nested leave-one-subject-out (LOSO) evaluation, are deliberately standard, so that the budget measures the pipeline rather than a new model; what is new is that every stage reports what it was worth.

The case study is jaw activity from surface EMG. Bruxism, habitual clenching or grinding of the teeth, is common and hard to observe directly [6], [7]. Surface EMG records the jaw-closing muscles directly [8], [9], but clenching, grinding and chewing all contract the same masseter and temporalis, and separating them on an unseen person is hard, yet good accuracies have been reported without an account of their origin [10]–[13]. Prior studies (Supplementary Table S-I) use small, mostly simulating cohorts and pooled validation, and vary only the classifier; we found none that reports a measured statement about the signal entering its pipeline, or that prices its segmentation and normalisation choices against its model. No public dataset supports such an analysis, so the data were collected for this study (Section II-A).

Four questions are posed. **RQ1:** how much of what a classifier learns from these recordings is physiology, and how much is mains interference? **RQ2:** can four EMG channels separate quiet rest from four instructed jaw activities on an unseen participant? **RQ3:** which part of the model does the work: the wavelet representation, the architecture built on it, or the number of parameters? **RQ4:** how do those modelling choices compare with choices fixed before any model was trained: window length, temporal context and normalisation scope?

Signal quality sets a ceiling; below it, what a model can represent matters more than its size; and choices made before training were worth as much as the architecture or more. The contributions are the measured budget (Table IV); the

Manuscript submitted September 7, 2026. This work was supported by an award from the Rhode Island Life Science Hub Small Grant Program (Principal Investigator T. A. Walls; Co-Investigators R. Abiri and M. P. Furmanek) and in part by the U.S. National Science Foundation under Grant 2245558 (Principal Investigator R. Abiri).

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purpose-built dataset; a five-class result on unseen participants that includes quiet rest; a pipeline regenerated from a saved prediction ledger; and two inexpensive checks, drawing the filter response over the measured spectrum and fingerprinting every channel of every recording.

II. MATERIALS AND METHODS

A. Data

The study was approved by the Institutional Review Board of the University of Rhode Island (Protocol #IRB22275690-2); every participant gave written informed consent, including consent to publish the identifying image in Fig. 1(a). The recordings were collected for this study because no public dataset offers what the analysis needs: continuous recordings archived as acquired with a record of the hardware's own filtering, a synchronised marker, a quiet non-event condition, more than one food [14], and raw files from which channel provenance can be verified (Supplementary Section S-II).

B. Participants, Procedure and Sensors

Five adults (two women, three men; age 38.0 ± 10.4 years) with a prior clinical diagnosis of bruxism performed every task on instruction, awake, in the laboratory: quiet rest; jaw movements (opening and closing, lateral deviation, protrusion and retrusion, biting); isometric molar and incisor clenching; instructed grinding; and chewing of cheese, carrots or popcorn, and gum, in twenty 1-min single-condition recordings per participant. A handheld button, pressed by the participant at task initiation, transmitted synchronised markers that are the sole source of the class labels. Four differential surface-EMG channels were recorded from bilateral bipolar pairs over the masseter and temporalis (Fig. 1(a)); a camera served offline review, and a microphone, also recorded, enters only the RQ1 contrast (Table I). Signals were sampled at 1200 Hz and written to disk as acquired, with a metadata sidecar recording the acquisition software's own filter settings. Amplitudes are in arbitrary converter units, the hardware not being documented to current standards [15], [16].

C. Windows and Labels

Every analysed window lies wholly inside one marked task interval and is cut from the continuously filtered recording. Three measured exclusions apply (Supplementary Table S-II): a 0.25-s guard at every marker edge and a 0.5-s startup guard, both sized from measured onset lags and settling transients, and quiet-rest windows drawn only from the five dedicated rest recordings. Five classes result: rest; movement; clenching (molar, incisor and unilateral bite); grinding; and chewing. Ninety-nine recordings yielded 6,173 analysable 1.0-s windows at a 0.5-s stride, 58.9% of them chewing. The participant, not the window, is the unit of generalisation [17].

D. Signal Validation and the Filter Chain

The acquisition software applied its own fixed notch, recorded in every sidecar, which removed the 60-Hz fundamental before the data reached disk; interference survived at

the harmonics, 180, 300 and 420 Hz, holding 91.3–99.8% of the quiet-rest recordings' raw in-band power. The earlier chain (hereafter the superseded chain), a single 60-Hz notch ($Q = 30$) followed by a 20–450-Hz band-pass, therefore removed a frequency that was already absent and passed every one that was present (Fig. 2(a), (c)).

The corrected chain notches every mains multiple from 60 to 420 Hz before the same fourth-order Butterworth band-pass from 20 to 450 Hz [18], [19]. Each notch is 8 Hz wide at -3 dB rather than constant- Q : at an identical 13% band cost the constant-width bank left eight times less residual interference, and the quiet-rest mains share fell to 0.46–0.66% (Fig. 2(b), (c)). The chain was selected on this criterion, not on held-out score (Supplementary Section S-III). Harmonic notching is standard [20], [21]; the novelty claimed is the check, not the remedy.

E. Normalisation, Wavelet Decomposition and Model

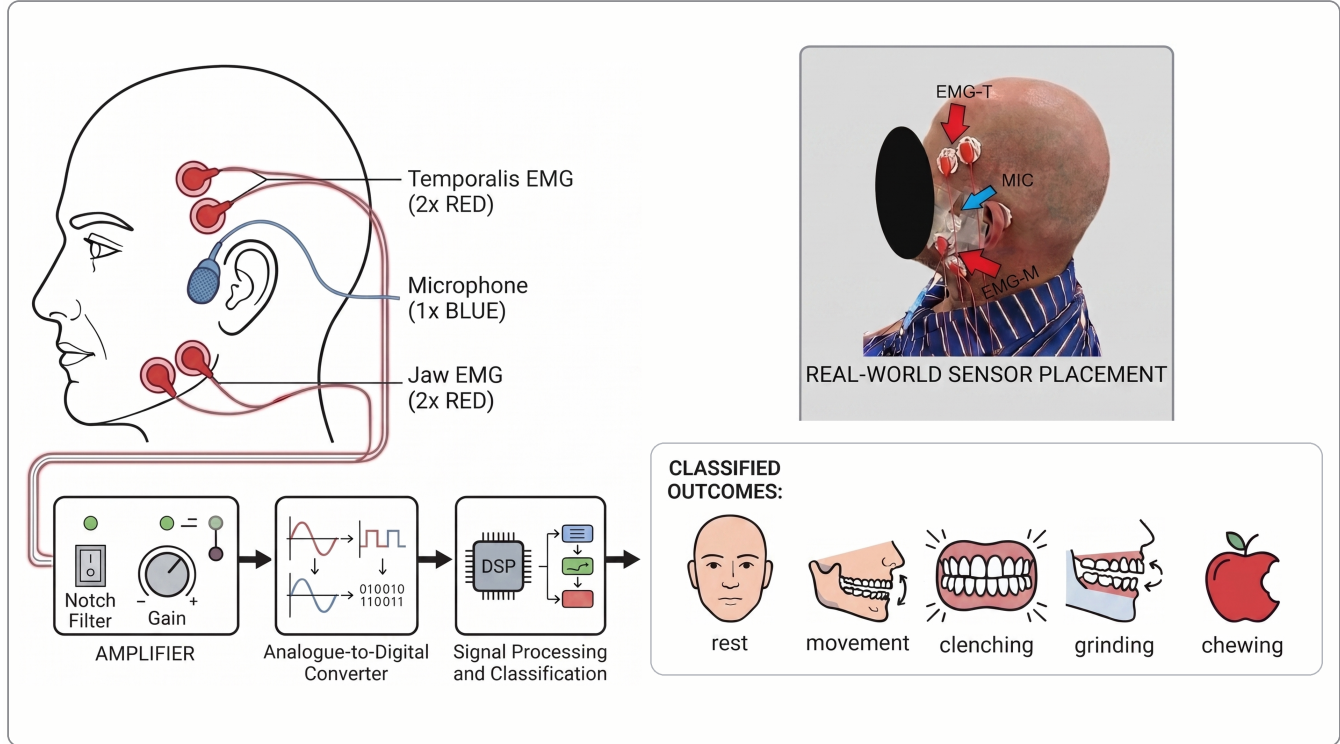
Channels were z-scored with statistics computed only from the training participants of the fold in question. Signals were decomposed with the Daubechies-4 wavelet at four levels [22], [23], retaining A_4 (below 38 Hz), D_3 (75–150 Hz) and D_1 (300–450 Hz after the band-pass). The reference model (Fig. 1(b)) gives each band two 3-tap convolutional blocks of 8 and 16 channels with batch normalisation and rectified linear units, averaged over the window's time axis to 16 features per band; the 48 features pass through a multilayer perceptron ($48 \rightarrow 48 \rightarrow 32$, dropout 0.5) to five logits, 5,901 trainable parameters in all.

F. Training and Evaluation

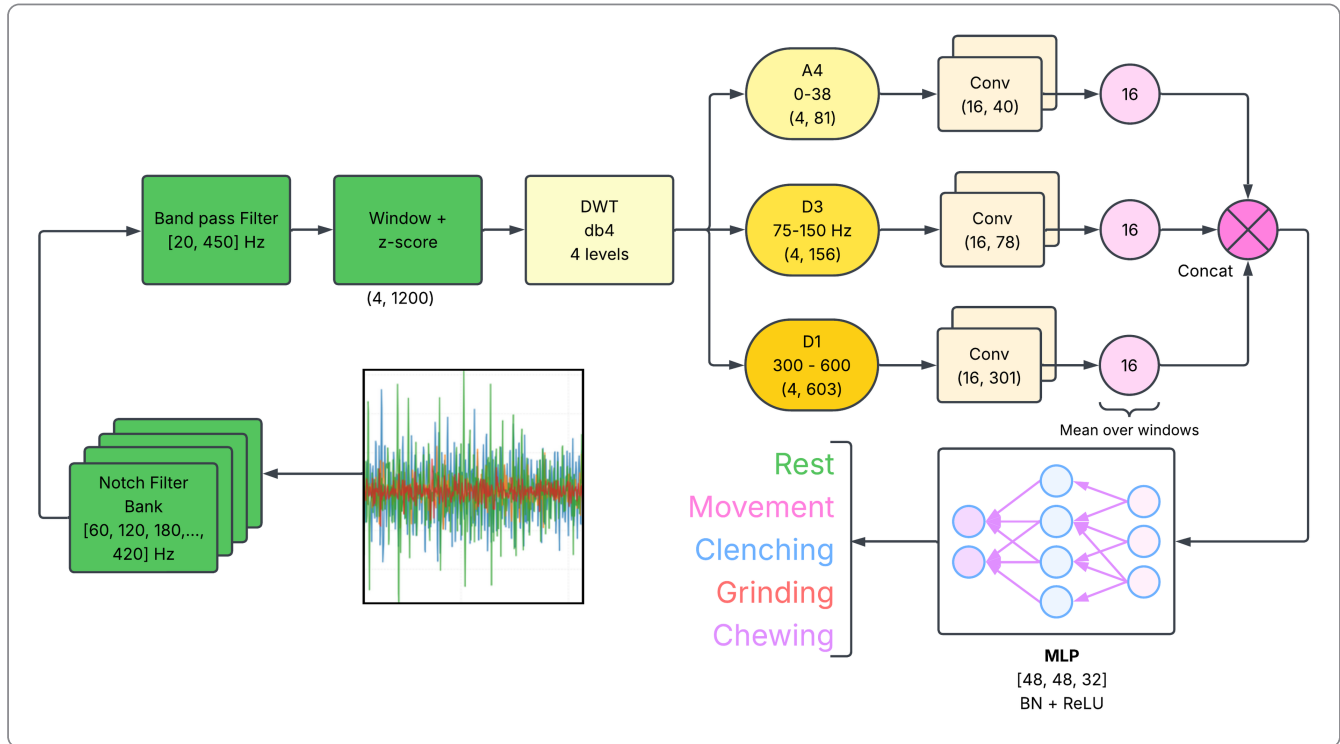
Training used inverse-frequency class weights and minority-class augmentation, both confined to each fold's training partition, and AdamW [24] with focal loss [25] in PyTorch [26]; the loss and learning rate were fixed rather than searched (Supplementary Table S-II). Evaluation was nested LOSO [27], [28]: in each of five outer folds one participant was sealed as the test set, every statistic, weight, augmentation set and epoch budget was fitted inside a participant-grouped inner loop on the remaining four, and the held-out participant was scored exactly once; three seeds were run, none selected as best. Participant-level metrics, computed within each held-out participant and then averaged, are primary; pooled-window metrics are descriptive only, and no window-level intervals or p -values are reported, because the windows are clustered and $N = 5$.

G. Screening Harness and Safeguards

A deliberately cheap screening harness supplies measurements the confirmatory protocol could not afford: 28 hand-crafted band-energy and waveform features per window [29], with logistic regression and gradient boosting [30] fitted once per outer fold. Because the features were chosen after seeing these participants, screening numbers are reliable in direction and optimistic in level. Three guards inside the pipeline refuse rather than warn: a leakage assertion on normalisation statistics,



(a)



(b)

Fig. 1. System overview and reference model. (a) Acquisition chain: bipolar surface-EMG pairs over the temporalis and the masseter (“jaw”) on each side, four channels in all; the amplifier and acquisition software, whose own fixed notch is recorded in every metadata sidcar; the analogue-to-digital converter; and the processing that assigns each 1.0-s window to quiet rest or one of four instructed jaw activities. The microphone (blue) was recorded alongside and enters only the RQ1 contrast of Table I. Inset: placement on a participant (EMG-T, temporalis; EMG-M, masseter; MIC, microphone). (b) Reference model: the four channels pass through the notch bank and band-pass, are cut into 1.0-s windows, z-scored with training-participant statistics and decomposed with a four-level db4 transform; the retained bands A_4 , D_3 and D_1 (D_1 occupies 300–450 Hz after the band-pass) each pass through two convolutional blocks and are averaged over the window’s time axis to 16 features, and the 48 features enter an MLP (48 → 48 → 32) that produces five logits. The extended model of Table III keeps the time axis instead of averaging it.

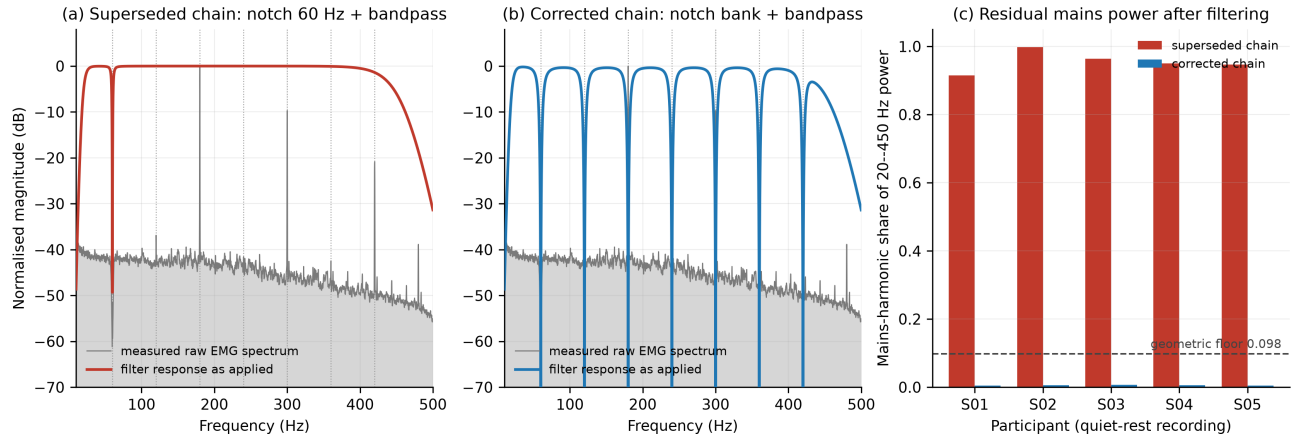


Fig. 2. The mains-harmonic defect and its correction. (a) Superseded chain: the 60-Hz notch lands where the hardware had already removed the fundamental, while the harmonics at 180, 300 and 420 Hz pass untouched. (b) Corrected chain: seven constant-width notches land on the peaks. Gray traces are the mean raw spectrum of the five quiet-rest recordings, colored traces the response as applied, dotted lines the mains multiples. (c) Share of 20–450-Hz power within ± 3 Hz of a mains multiple, per participant, after each chain; the dashed line marks the 0.098 a white signal would score.

a check against wavelet bands the filter chain has removed, and a channel fingerprint against any held-out participant's signal appearing in its training set.

III. RESULTS

The confirmatory results come from one nested-LOSO run on the corrected signal, 18,519 held-out predictions over five folds and three seeds, every value recomputed from the saved prediction ledger; pre-correction results appear only as RQ1's contrast arm.

A. RQ1: How Much of the Signal Is Physiology

The filter chain was changed while the windows, folds and labels were held identical (Table I). Through the superseded chain the confirmatory network reached 55.0% accuracy, 43.5% participant-level macro-F1 and a macro AUC of 0.785; through the corrected chain the matched condition reached 80.9%, 72.8% and 0.963, a gain of 29.3 points of macro-F1, and the share of quiet-rest windows misread as clenching or grinding fell from 30.2% to 6.1%. The contrast is conservative: both arms are the EMG-plus-microphone condition, and the superseded arm additionally searched its loss and learning rate. The defect presented as between-participant heterogeneity: through the superseded chain P1 and P2 scored 18.1% and 5.4% macro-F1, at or below a constant prediction, whereas after correction all five participants span 61.4–82.4%. A control chain with no mains removal scored within 1.2 points of the superseded chain, which was therefore inert (Supplementary Fig. S2).

B. RQ2: Five Classes on Unseen Participants

Averaged over the five held-out participants and three seeds, the reference model achieved 81.7% accuracy, 72.0% macro-F1 and a macro ROC-AUC of 0.958 (Table II). Quiet rest reached 77.1% precision, 91.4% recall and an F1 of 83.6%, the best-recalled class. Movement's F1 of 46.0% is a precision

problem (Fig. 3), 34.4% precision at 69.8% recall, because the smallest class absorbs 9.9% of the eleven-times-more-numerous chewing windows [31]. Clenching and grinding are the hardest pair, 19.5% of clenching windows labelled grinding and 14.5% of grinding windows clenching. Rest's dominant error was into grinding, 7.5%, so 7.7% of quiet-rest windows fell on the tooth-contact side, the closest analogue of a false-alarm rate available here. Secondary analyses are in Supplementary Table S-III, ROC curves and a t-SNE in Supplementary Fig. S3.

C. RQ3: Representation, Architecture and Capacity

Logistic regression on the 28 screening features reached 63.9% macro-F1 and gradient boosting 61.3% (Table II); the network's 72.0% exceeds them by 8.1 and 10.6 points, a conservative margin given the harness's optimism. The two rank windows about equally well (macro AUC 0.965 against 0.958), so the network's advantage lies in where its decision boundaries fall; before the filter correction it had been only about four points, because when interference dominates a window a pooled band amplitude is all there is to extract (Supplementary Section S-IV).

A prespecified sweep then contrasts four architectures under one training configuration on identical inputs (Table III): the compact reference model; an extended member of the same family that keeps the time axis and adds a modulation-spectrum head, making 1–2-Hz rhythm representable within a window; and an early-fusion CNN and a bidirectional LSTM that read the raw channels without the wavelet decomposition. The grouping is the result: both wavelet models place above both raw-signal models, and the ordering does not follow parameter count, the early-fusion CNN reaching 68.9% macro-F1 with 3.2 times the compact model's parameters against its 72.2%, and the LSTM 60.2%. Within the wavelet family the extended model is stronger, 75.0% macro-F1 and 0.974 AUC against 72.2% and 0.954, consistently across seeds but on three of the five participants, at three times the parameters and 2.2 times the

TABLE I. RQ1. Only the filter chain varies: the same 6,173 windows, fold assignments and labels in every arm. Residual mains is the share of 20–450-Hz power within ± 3 Hz of a mains multiple (a white signal scores 0.098); “Inv.” counts participant pairs in which one person’s resting EMG exceeded another’s active EMG.

EMG chain	Measured signal properties			Participant-level macro-F1		
	Residual mains	Ampl. spread	Inv.	Wavelet CNN ^a	Logistic reg. ^b	Grad. boost. ^b
60-Hz notch only (superseded)	0.673	5.07×	20	43.5%	37.3%	38.8%
Band-pass only (control)	0.671	5.02×	20	—	38.5%	39.2%
Mains notch bank, 8 Hz (used)	0.012	2.97×	0	72.8%	63.9%	61.3%

^aConfirmatory nested LOSO, three seeds; both rows the EMG-plus-microphone condition, the superseded row additionally searching loss and learning rate, so the comparison favours the superseded arm. The corrected EMG-only condition scores 72.0%. ^bScreening harness, one fit per outer fold, no nested selection; optimistic in level.

TABLE II. Five-class performance on the corrected signal: participant-level means over five held-out participants and three seeds (upper block); per-class values on pooled held-out windows, averaged over seeds (middle); and each held-out participant, averaged over seeds (lower). Values describe this five-participant dataset and are not population-level tests.

Method / class / participant	Accuracy	Precision	Recall	F1	AUC	Windows
<i>Participant-level means, five held-out participants $\times$ three seeds</i>						
Wavelet CNN, 4-channel EMG	81.7%	73.7%	77.6%	72.0%	0.958	6,173
Logistic regression, band energies ^a	78.3%	—	—	63.9%	0.965	6,173
Gradient boosting, band energies ^a	74.8%	—	—	61.3%	0.914	6,173
<i>Wavelet CNN, per class, pooled held-out windows</i>						
Rest	—	77.1%	91.4%	83.6%	0.958	590
Movement	—	34.4%	69.8%	46.0%	0.937	333
Clenching	—	70.5%	66.0%	68.1%	0.936	799
Grinding	—	67.9%	69.5%	68.6%	0.928	816
Chewing	—	97.3%	85.8%	91.1%	0.981	3,635
Macro average	80.7%	69.4%	76.5%	71.5%	0.948	6,173
<i>Wavelet CNN, per held-out participant, three-seed means</i>						
Participant 1	73.4%	—	—	63.2%	0.904	1,692
Participant 2	90.7%	—	—	72.6%	0.957	1,007
Participant 3	83.7%	—	—	71.3%	0.967	1,133
Participant 4	74.0%	—	—	72.2%	0.981	1,167
Participant 5	86.7%	—	—	80.5%	0.980	1,174
Standard deviation across participants	7.7	—	—	6.1	0.032	—

^aScreening harness (Section II-G), optimistic in level, which makes the network’s margin conservative; the harness stores no per-class table.

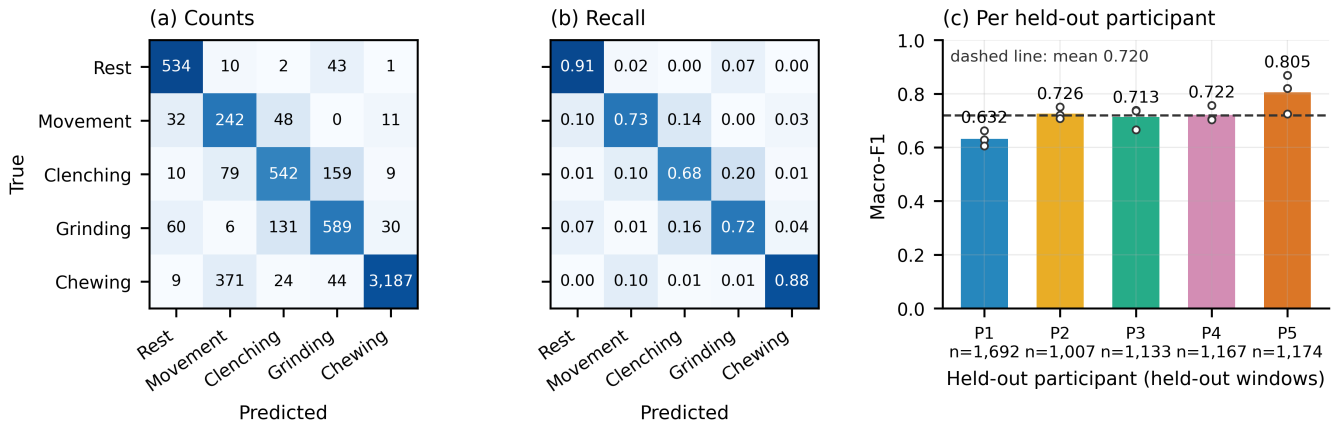


Fig. 3. Held-out behaviour of the reference model. (a) Confusion counts and (b) row-normalised recall, pooled across the five LOSO folds of seed 0, chosen by the fixed rule of lowest seed index. (c) Macro-F1 per held-out participant: bars are means over three seeds, open markers the individual seeds, and the label inside each bar that participant’s held-out window count.

latency. The compact model remains the reference: a 2.8-point difference resting on three of five participants is a direction rather than an effect size, it delivers 96% of the extended model’s macro-F1 at 33% of the parameters, and the two are one family, so the extended model’s advantage supports the

representational argument.

D. RQ4: Three Choices Fixed Before Modelling

Three design choices were priced with the screening harness (Fig. 4); two are worth as much as the 8.1-point architectural

TABLE III. RQ3. Four architectures on identical inputs, windows, folds and seeds under one training configuration, grouped by front end. Participant-level means $\pm$ standard deviation across seeds; “Forward” is the median forward pass for one window at batch size 1 on an Intel CPU.

Architecture	Params	Forward	Accuracy	Bal. acc.	Macro-F1	Macro AUC
<i>db4 wavelet front end</i>						
Wavelet CNN, extended ^a	17,981	2.21 ms	86.1 $\pm$ 1.1%	78.9 $\pm$ 0.9%	75.0 $\pm$ 1.3%	0.974 $\pm$ 0.003
Wavelet CNN, compact (reference) ^b	5,901	1.02 ms	81.0 $\pm$ 2.2%	77.4 $\pm$ 1.0%	72.2 $\pm$ 1.5%	0.954 $\pm$ 0.006
<i>No wavelet decomposition</i>						
Early-fusion CNN, raw signals	19,029	0.20 ms	79.0 $\pm$ 0.8%	74.3 $\pm$ 1.3%	68.9 $\pm$ 1.5%	0.949 $\pm$ 0.004
Bidirectional LSTM	10,053	1.16 ms	71.5 $\pm$ 3.3%	66.4 $\pm$ 2.5%	60.2 $\pm$ 2.6%	0.898 $\pm$ 0.013

^aThe reference model with the time axis kept, plus mean, standard-deviation and maximum pooling per band and a modulation-spectrum head; its advantage rests on three of the five participants. ^bThe sweep’s own measurement of the reference model; the confirmatory run reached 72.0% (Table II) on a different code revision.

margin or more. First, the 1.0-s window is a binding constraint, not a tuned parameter: chewing and grinding differ from clenching partly by 1–2-Hz rhythm (Supplementary Fig. S4), and aggregating held-out probabilities over consecutive windows raises macro-F1 monotonically from 0.730 at 1.0 s to 0.804 at 5.5 s. Because every recording holds one condition, that 7.4-point gain is a trial-level estimate, not a streaming result. Second, because participants marked intervals at very different granularity, the guard sweep alone moves the smallest participant-and-class cell from 50 windows to 1; the resolution is to standardise trigger use at collection. Third, normalisation scope is worth about as much as the architecture: standardising each participant’s features by that participant’s own unlabelled distribution raises the screening classifier from 0.639 to 0.730 macro-F1, transductive adaptation that is legitimate but presumes a per-user fitting session, so it is used by no reported number. Training curves and latencies are in Supplementary Fig. S5; no real-time claim is made.

E. What Determines Performance

Table IV collects the six choices on one scale; the ordering is the paper’s main result. Signal quality sets a ceiling that nothing above it can raise: through the superseded chain the best model scored 43.5% macro-F1, below the 61.3% the weakest baseline reaches on the corrected signal. The entries are not additive (Table IV, note ^e), three are screening measurements, optimistic in level, and the magnitudes belong to this cohort; what generalises is the ordering.

IV. DISCUSSION

A. What the Ordering Means

What a model could represent mattered more than how large it was, and signal quality set the ceiling for both. Within the wavelet pair, the model that could also represent rhythm beat the one that could not, which converges with RQ4: the rhythm the extended model represents is the information the 1.0-s window withholds, worth 7.4 points against the architecture’s 2.8. The binding limitation is informational rather than architectural: a longer window would likely gain more than a larger model. RQ1 is easy to overstate: mains correction is not generally worth 29.3 points, and with clean recordings it is worth nothing; the claim is that the ceiling exists, that its height

is measurable, and that nothing above it can be interpreted until it has been measured.

B. Why Signal Validation Belongs Inside the Pipeline

The defect did not look like a signal problem. It looked like between-participant heterogeneity, the failure mode that motivates transfer learning and larger cohorts, because interference amplitude varies with electrode impedance, cable routing and session rather than with physiology [16], [19], LOSO attributes that variation to the person held out, and the result is the distribution shift that domain generalisation sets out to absorb [32]. The code implemented the recommended filter [8], [33]; what was missing was one comparison, the response drawn over the measured spectrum of the recordings, and one statistic, the share of in-band power near each mains multiple. That is the argument for making validation a stage of the pipeline that can refuse.

C. What Transfers Beyond This Case Study

Nothing in the procedure is specific to EMG or to bruxism. Nor is it an ablation study, which varies parts of a model; the budget varies the stages upstream of the model, on identical windows and folds. Audit the signal against what the analysis assumes and publish the resulting statistics beside the accuracy; fix the windows, folds and labels and vary one stage at a time; include the choices made before any model was trained; report a budget beside the accuracy, naming each row’s harness and baseline; and keep a prediction ledger from which every number is regenerated. EEG, where preprocessing choices are contested [34] and architectures are compared under one unpriced pipeline [3], is the obvious next case. What does not transfer is the numbers: clean mains would make the first row of Table IV worth nothing, a modality without rhythmic content would make temporal context worth less, and balanced classes would make normalisation scope worth less.

D. Limitations and Next Steps

The principal limitation is the cohort size, $N = 5$: 6,173 overlapping windows do not replace independent participants, one of whom contributed 27% of them, and LOSO does not establish external generalisation from five people [35]. The budget is an ordering with approximate magnitudes, not a

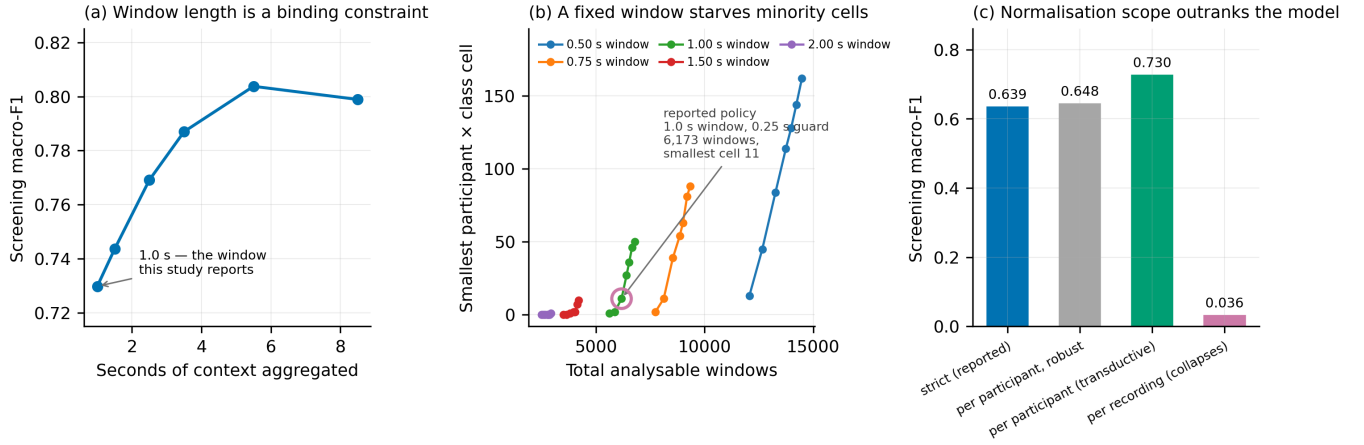


Fig. 4. Three design constraints, measured on identical windows and folds with the screening harness. (a) Macro-F1 against seconds of temporal context aggregated within a recording. (b) Every window and guard setting swept: total analysable windows against the smallest participant-and-class cell, reported policy circled. (c) Normalisation scope; the per-participant scope is used by no reported number.

TABLE IV. What determines performance: six choices priced on the same 6,173 windows and five outer folds, in participant-level macro-F1. Entries are not additive and not all from the same harness (Section III-E). “Reported” states whether the more favourable setting is the one used in this paper.

Choice	Change measured	Harness	Δ macro-F1	Reported
Signal quality	60-Hz notch $\rightarrow$ harmonic notch bank	Confirmatory ^a	+29.3	Yes
Normalisation scope	Strict $\rightarrow$ per-participant	Screening ^b	+9.1	No ^c
Learned representation	Band energies $\rightarrow$ wavelet CNN	Conf./screen. ^d	+8.1	Yes
Temporal context	1.0 s $\rightarrow$ 5.5 s within a trial	Screening ^{b,e}	+7.4	No ^e
Rhythm representation	Compact $\rightarrow$ extended wavelet CNN	Arch. sweep	+2.8	Secondary
Capacity, raw signal	Wavelet CNN $\rightarrow$ early-fusion (3.2 $\times$)	Arch. sweep	-3.3	No

^aNested LOSO, three seeds, both arms the EMG-plus-microphone condition (Table I). ^bScreening harness: optimistic in level, reliable in direction.

^cLegitimate, but presumes a per-user fitting session. ^dConfirmatory network against a screening baseline, the conservative direction. ^eTrial-level: every recording holds one condition. Its 1.0-s baseline is the per-participant scope of the second row, so the two rows share a baseline and must not be summed.

decomposition of variance, and the RQ1 contrast, although conservative in protocol, compares runs from two codebase versions on verified byte-identical windows. The classes are instructed laboratory conditions, not the consensus definition of bruxism [36]: the tasks omit bracing and thrusting [37], the diagnoses were not phenotyped [38], labels rest on a button press, instructed activity may be more regular than spontaneous activity [39], and rest did not sample speaking, swallowing or head movement, so the analysis estimates confusion against quiet rest, not false alarms per hour; pooled-validation accuracies above 90% [10] are not like-for-like. Chewing carries the headline accuracy, and part of the clenching-and-grinding confusion is label ambiguity, since a grinding bout’s static phases may legitimately carry either label [40]. A next study should aggregate over a declared temporal context, standardise trigger use, and record over several days with a defined background class; nothing about sleep follows from these awake tasks [7], [41].

V. CONCLUSION

This study asked which stages of one biosignal classification pipeline account for its performance, and answered by measurement. On unseen participants the pipeline separated quiet rest and four instructed jaw activities at 81.7% accuracy and 72.0% macro-F1 with 5,901 parameters. The ordering is

the more durable result: signal quality set a ceiling, a mains-harmonic defect the conventional chain had left untouched being worth 29.3 points of macro-F1; above it, what a model could represent mattered more than its size; and two choices fixed before any model was trained, temporal context and normalisation scope, were each worth as much as the architecture or more. The transferable recommendation costs one pass over the data: report a measured interference statistic beside every accuracy, draw filter responses over the measured spectrum, and fingerprint every channel of every recording. This study does not validate a bruxism detection device; it shows that surface EMG with a wavelet representation merits a larger, phenotyped, multi-day evaluation, and that between any recording and a reported accuracy sit half a dozen stages, each of which can be worth more than the model and each of which can be priced with the data already in hand.

SUPPLEMENTARY MATERIALS

The Supplementary Materials contain the positioning comparison with prior studies (Table S-I); the dataset, guard and training configuration (Table S-II); the secondary analyses (Table S-III); the preprocessing chain on a real excerpt (Fig. S1); the RQ1 mechanism in signal terms (Fig. S2); per-class ROC curves and a t-SNE of the learned representation (Fig. S3); signal excerpts with the 1.0-s window drawn to scale (Fig. S4);

training curves (Fig. S5); and the data, code and artifact-provenance statements.

ACKNOWLEDGMENT

The authors thank University of Rhode Island students Anjolaoluwa Akingbile, Jessica Constantine and Nathan DeSalvo, who located and summarised the functions of bruxism devices before this project. The authors declare no competing interests.

REFERENCES

- [1] M. Thymi, F. Lobbezoo, G. Aarab, J. Ahlberg, K. Baba, M. C. Carra, L. M. Gallo, A. De Laat, D. Manfredini, G. Lavigne, and P. Svensson, "Signal acquisition and analysis of ambulatory electromyographic recordings for the assessment of sleep bruxism: A scoping review," *Journal of Oral Rehabilitation*, vol. 48, no. 7, pp. 846–871, 2021.
- [2] Y. Roy, H. Banville, I. Albuquerque, A. Gramfort, T. H. Falk, and J. Faubert, "Deep learning-based electroencephalography analysis: a systematic review," *Journal of Neural Engineering*, vol. 16, no. 5, p. 051001, 2019.
- [3] X. Bouthillier, P. Delaunay, M. Bronzi, A. Trofimov, B. Nichyporuk, J. Szeto, N. Mohammadi Sepahvand, E. Raff, K. Madan, V. Voleti, S. Ebrahimi Kahou, V. Michalski, T. Arbel, C. Pal, G. Varoquaux, and P. Vincent, "Accounting for variance in machine learning benchmarks," in *Proceedings of Machine Learning and Systems (MLSys)*, vol. 3, 2021, pp. 747–769.
- [4] S. Kapoor, E. M. Cantrell, K. Peng, T. H. Pham, C. A. Bail, O. E. Gundersen, J. M. Hofman, J. Hullman, M. A. Lones, M. M. Malik, P. Nanayakkara, R. A. Poldrack, I. D. Raji, M. Roberts, M. J. Salganik, M. Serra-Garcia, B. M. Stewart, G. Vandewiele, and A. Narayanan, "REFORMS: Consensus-based recommendations for machine-learning-based science," *Science Advances*, vol. 10, no. 18, p. eadk3452, 2024.
- [5] G. S. Collins, K. G. M. Moons, P. Dhiman, R. D. Riley, A. L. Beam, B. Van Calster, M. Ghassemi, X. Liu, J. B. Reitsma, M. van Smeden *et al.*, "TRIPOD+AI statement: Updated guidance for reporting clinical prediction models that use regression or machine learning methods," *BMJ*, vol. 385, p. e078378, 2024.
- [6] G. J. Lavigne, S. Khoury, S. Abe, T. Yamaguchi, and K. Raphael, "Bruxism physiology and pathology: An overview for clinicians," *Journal of Oral Rehabilitation*, vol. 35, no. 7, pp. 476–494, 2008.
- [7] G. J. Lavigne, P. H. Rompré, and J. Y. Montplaisir, "Sleep bruxism: Validity of clinical research diagnostic criteria in a controlled polysomnographic study," *Journal of Dental Research*, vol. 75, no. 1, pp. 546–552, 1996.
- [8] M. B. I. Reaz, M. S. Hussain, and F. Mohd-Yasin, "Techniques of EMG signal analysis: Detection, processing, classification and applications," *Biological Procedures Online*, vol. 8, no. 1, pp. 11–35, 2006.
- [9] A. Colonna, L. Noveri, M. Ferrari, A. Bracci, and D. Manfredini, "Electromyographic assessment of masseter muscle activity: A proposal for a 24 h recording device with preliminary data," *Journal of Clinical Medicine*, vol. 12, no. 1, p. 247, 2023.
- [10] T. Sonmezocak and S. Kurt, "Detection of EMG signals by neural networks using autoregression and wavelet entropy for bruxism diagnosis," *Elektronika ir Elektrotechnika*, vol. 27, no. 2, pp. 11–21, 2021.
- [11] J. Z. Gul, N. Fatima, Z. Mohy Ud Din, M. Khan, W. Y. Kim, and M. M. Rehman, "Advanced sensing system for sleep bruxism across multiple postures via EMG and machine learning," *Sensors*, vol. 24, no. 16, p. 5426, 2024.
- [12] A. Ishtiaq, J. Gul, Z. Mohy Ud Din, A. Imran, and K. El Hindi, "Wearable regionally trained AI-enabled bruxism-detection system," *IEEE Access*, vol. 13, pp. 15 503–15 528, 2025.
- [13] M. B. B. Heyat, F. Akhtar, A. Khan, A. Noor, B. Benjdira, Y. Qamar, S. J. Abbas, and D. Lai, "A novel hybrid machine learning classification for the detection of bruxism patients using physiological signals," *Applied Sciences*, vol. 10, no. 21, p. 7410, 2020.
- [14] R. Zhang and O. Amft, "Monitoring chewing and eating in free-living using smart eyeglasses," *IEEE Journal of Biomedical and Health Informatics*, vol. 22, no. 1, pp. 23–32, 2018.
- [15] International Society of Electrophysiology and Kinesiology, "Standards for reporting EMG data," ISEK standards document, prepared by R. Merletti, 1999, <https://isek.org/emg-standards/>.
- [16] M. Besomi, P. W. Hodges, J. Van Dieën, R. G. Carson, E. A. Clancy, C. Disselhorst-Klug, A. Holobar, F. Hüg, M. C. Kiernan, M. Lowery *et al.*, "Consensus for experimental design in electromyography (CEDE) project: Electrode selection matrix," *Journal of Electromyography and Kinesiology*, vol. 48, pp. 128–144, 2019.
- [17] S. Saeb, L. Lonini, A. Jayaraman, D. C. Mohr, and K. P. Kording, "The need to approximate the use-case in clinical machine learning," *GigaScience*, vol. 6, no. 5, p. gix019, 2017.
- [18] A. van Boxtel, "Optimal signal bandwidth for the recording of surface EMG activity of facial, jaw, oral, and neck muscles," *Psychophysiology*, vol. 38, no. 1, pp. 22–34, 2001.
- [19] C. J. De Luca, L. D. Gilmore, M. Kuznetsov, and S. H. Roy, "Filtering the surface EMG signal: Movement artifact and baseline noise contamination," *Journal of Biomechanics*, vol. 43, no. 8, pp. 1573–1579, 2010.
- [20] D. T. Mewett, K. J. Reynolds, and H. Nazeran, "Reducing power line interference in digitised electromyogram recordings by spectrum interpolation," *Medical & Biological Engineering & Computing*, vol. 42, no. 4, pp. 524–531, 2004.
- [21] M. R. Keshkaran and Z. Yang, "A fast, robust algorithm for power line interference cancellation in neural recording," *Journal of Neural Engineering*, vol. 11, no. 2, p. 026017, 2014.
- [22] I. Daubechies, *Ten Lectures on Wavelets*. Philadelphia, PA: Society for Industrial and Applied Mathematics, 1992.
- [23] K. Englehart and B. Hudgins, "A robust, real-time control scheme for multifunction myoelectric control," *IEEE Transactions on Biomedical Engineering*, vol. 50, no. 7, pp. 848–854, 2003.
- [24] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *Proceedings of the 7th International Conference on Learning Representations (ICLR)*, New Orleans, LA, USA, 2019.
- [25] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *2017 IEEE International Conference on Computer Vision (ICCV)*, Venice, Italy, Oct. 2017, pp. 2999–3007.
- [26] A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein, L. Antiga *et al.*, "PyTorch: An imperative style, high-performance deep learning library," in *Advances in Neural Information Processing Systems 32 (NeurIPS 2019)*, Vancouver, BC, Canada, 2019, pp. 8024–8035.
- [27] G. C. Cawley and N. L. C. Talbot, "On over-fitting in model selection and subsequent selection bias in performance evaluation," *Journal of Machine Learning Research*, vol. 11, pp. 2079–2107, 2010.
- [28] S. Varma and R. Simon, "Bias in error estimation when using cross-validation for model selection," *BMC Bioinformatics*, vol. 7, p. 91, 2006.
- [29] A. Phinyomark, P. Phukpattaranont, and C. Limsakul, "Feature reduction and selection for EMG signal classification," *Expert Systems with Applications*, vol. 39, no. 8, pp. 7420–7431, 2012.
- [30] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg *et al.*, "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [31] T. Saito and M. Rehmsmeier, "The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets," *PLOS ONE*, vol. 10, no. 3, p. e0118432, 2015.
- [32] J. Wang, C. Lan, C. Liu, Y. Ouyang, T. Qin, W. Lu, Y. Chen, W. Zeng, and P. S. Yu, "Generalizing to unseen domains: A survey on domain generalization," *IEEE Transactions on Knowledge and Data Engineering*, vol. 35, no. 8, pp. 8052–8072, 2023.
- [33] R. H. Chowdhury, M. B. I. Reaz, M. A. B. M. Ali, A. A. A. Bakar, K. Chellappan, and T. G. Chang, "Surface electromyography signal processing and classification techniques," *Sensors*, vol. 13, no. 9, pp. 12 431–12 466, 2013.
- [34] A. Delorme, "EEG is better left alone," *Scientific Reports*, vol. 13, no. 1, p. 2372, 2023.
- [35] G. Varoquaux, "Cross-validation failure: Small sample sizes lead to large error bars," *NeuroImage*, vol. 180, pp. 68–77, 2018.
- [36] F. Lobbezoo, J. Ahlberg, K. G. Raphael, P. Wetselaar, A. G. Glaros, T. Kato, V. Santiago, E. Winocur, A. De Laat, R. De Leeuw *et al.*, "International consensus on the assessment of bruxism: Report of a work in progress," *Journal of Oral Rehabilitation*, vol. 45, no. 11, pp. 837–844, 2018.
- [37] M. C. Verhoeff, F. Lobbezoo, J. Ahlberg, S. Bender, A. Bracci, A. Colonna, C. Dal Fabbro, J. Durham, A. G. Glaros, B. Häggman-Henrikson *et al.*, "Updating the bruxism definitions: Report of an international consensus meeting," *Journal of Oral Rehabilitation*, vol. 52, no. 9, pp. 1335–1342, 2025.
- [38] D. Manfredini, J. Ahlberg, G. Aarab, S. Bender, A. Bracci, P. A. Cistulli, P. C. Conti, R. De Leeuw, J. Durham, A. Emodi-Perlman *et al.*,

- “Standardised tool for the assessment of bruxism,” *Journal of Oral Rehabilitation*, vol. 51, no. 1, pp. 29–58, 2024.
- [39] M. Farella, S. Palla, S. Erni, A. Michelotti, and L. M. Gallo, “Masticatory muscle activity during deliberately performed oral tasks,” *Physiological measurement*, vol. 29, no. 12, pp. 1397–1410, 2008.
- [40] K.-W. Lan, L.-L. Jiang, and Y. Yan, “Comparative study of surface electromyography of masticatory muscles in patients with different types of bruxism,” *World Journal of Clinical Cases*, vol. 10, no. 20, pp. 6876–6889, 2022.
- [41] E. Casett, J. C. Reus, J. Stuginski-Barbosa, A. L. Porporatti, M. C. Carra, M. A. Peres, G. de Luca Canto, and D. Manfredini, “Validity of different tools to assess sleep bruxism: A meta-analysis,” *Journal of Oral Rehabilitation*, vol. 44, no. 9, pp. 722–734, 2017.